ability enables WEAVER to mine training data for itself on large-scale corpus, opening the possibility of *scalable self-training on web data*.

3.1.3. Evaluation (Literary Critic)

Many recent work explored using or training LLMs to evaluate general instruction following tasks (Chan et al., 2023; Jiang et al., 2023b; Wang et al., 2023b). However, we find generalist LLMs require extensive prompting skills to make them suitable for evaluating tasks related to creative writing. Moreover, since almost all students majoring in creative writing are also required to take literary critic courses, we think learning to perform literary critic may be helpful for the model to produce better texts as well. Therefore, we also train WEAVER to judge the quality of the responses to writing instructions and do pairwise comparison of two responses.

We collect human preference between model outputs in WAWAWRITER, our AI-assisted writing platform and convert the collected preference data to training data for LLM-based evaluation with carefully curated templates.

3.1.4. Retrieval-Augmented Generation

The ability of retrieval-augmented generation (RAG) (Gao et al., 2023; Lewis et al., 2020), i.e., generating responses by referring to external knowledge or references as context. RAG is an important technique that helps LLMs generate more accurate and informed responses. It can be especially helpful for writing purposes since it's common for human writers to refer to other text samples when writing fiction or articles. However, most existing LLMs purely rely on prompt engineering to do RAG and do not perform RAG training during alignment. We believe this limits the ability of LLMs to make use of retrieved contexts. Therefore, we propose to include RAG-aware training data during alignment to enhance WEAVER's retrieval-augmented generation ability. Specifically, we augment 10% percent of training data by appending a relevant context obtained by retrieving the paragraph most similar to the target response. In this way, WEAVER learns to write by referring to external contexts and is thus more compatible with RAG techniques compared to most existing LLMs.

3.1.5. Function Calling

The ability to use tools is also very important for LLMs (Schick et al., 2023). This ability, also referred to as "function calling", is also helpful for writing because the model may need to search the internet for references or call editor APIs when doing human-AI collaborative writing. To unlock the function calling ability, we include an open-source function calling dataset[2] into supervised fine-tuning data. We also propose a new pipeline to synthesize more diverse function calling data by first using GPT-4 to synthesize diverse environments with multiple tools and APIs, as well as their documentation. We then randomly select one API at a time and ask GPT-4 to imagine a situation where the API can be helpful and the plausible arguments for the API. We then reason what one may instruct an LLM in that situation so that the API should be used with the arguments. Finally, similar to how GPTs support function calling, we train WEAVER to use tools by selecting the right API and generating the arguments given the instructions and the contexts.

3.2. Instruction Backtranslation

We then describe our proposed improved pipeline for instruction backtranslation. The motivation for doing instruction backtranslation instead of instruction augmentation methods such as self-

[2]https://huggingface.co/glaiveai

Domain	Subdomain	Description	Source
Fiction Writing	**Full Novel**	Web novel, over 1M words	Proprietary
	Short Story	Web stories, 10k-20k words	Proprietary
Creative Non-Fiction Writing	**Red**	Top liked and commented posts on Red	Picked
	Zhihu	Top upvoted posts on Zhihu	Picked
	Weibo	Top liked posts on Weibo	Picked
	WeChat Articles	Top read articles on WeChat	Picked
	DouBan	Top liked posts on DouBan	Picked
	News & Blogs	Popular news/blogs	Picked
Technical Writing	**Papers**	Academic papers on CNKI	Picked
	Essay	Online essays	Picked
	Contract	Contracts from online sources	Picked
	Reports	Reports for work, business, science, etc.	Proprietary
	Copies	Business & Government copies	Proprietary
Marketing Writing	**Business Plans**	Business plans for projects and startups	Proprietary
	Industry Report	Research report for different industries	Proprietary
	Advertising Copy	Popular copies for advertisements	Picked
	Marketing Plan	Marketing plans for products & services	Picked
	Product Overview	Articles advertising products	Picked

Table 2 | Description of SFT Data sources. We combine similar subdomains in the same fields for simplicity. The entire training set covers 34 subdomains and around 500,000 instruction-output pairs. "Picked" means the raw data in the corresponding domains are manually selected.

instruct (Wang et al., 2023a) is very simple: we want to align WEAVER on high-quality, stylish, and

human-written texts. To achieve this goal, we first collect high-quality stories, fiction chapters, and copies of different domains. We list the categories of collected texts in Table 2.

We then use a carefully designed few-shot prompt template to synthesize instruction-response pairs for all aforementioned writing tasks. Specifically, for each subdomain-task pair, we annotate 5 cases of how one can write an instruction-response pair, including both the annotated results and the rationales for the annotation process: we first select a text span from a case as the output (except for outlining, brainstorming, and reviewing tasks where the output is transformed from the selected text span with an additional prompt). We then identify or produce the context for the output. For example, for the polishing task, the context should be a worse version of the target output, so we can modify the wording and structure of the target output to make it look worse. Then we infer the instruction that one may use to transform the context to the output. Taking the polishing task as an example again, we need to reason what modifications are made and synthesize the polishing instructions accordingly. For each unlabeled case, we use the annotated cases as few-shot exemplars and ask GPT-4 to first generate the annotation process in the Chain-of-Thought style (Wei et al., 2022) and then produce the synthesized instruction-response pairs. The instruction backtranslation pipeline is illustrated in Figure 1. We synthesize 500,000 high-quality instruction-response pairs across all domains and tasks with this pipeline. Finally, we do an instruction data selection procedure following the practice described in (Liu et al., 2023): we first score all instruction-response pairs with GPT-3.5-turbo and then select top-ranked data in each subdomain-task pair for supervised fine-tuning. Specifically, we score each instruction-response pair based on the quality and the diversity of the instruction and the relevance between the instruction and the response.

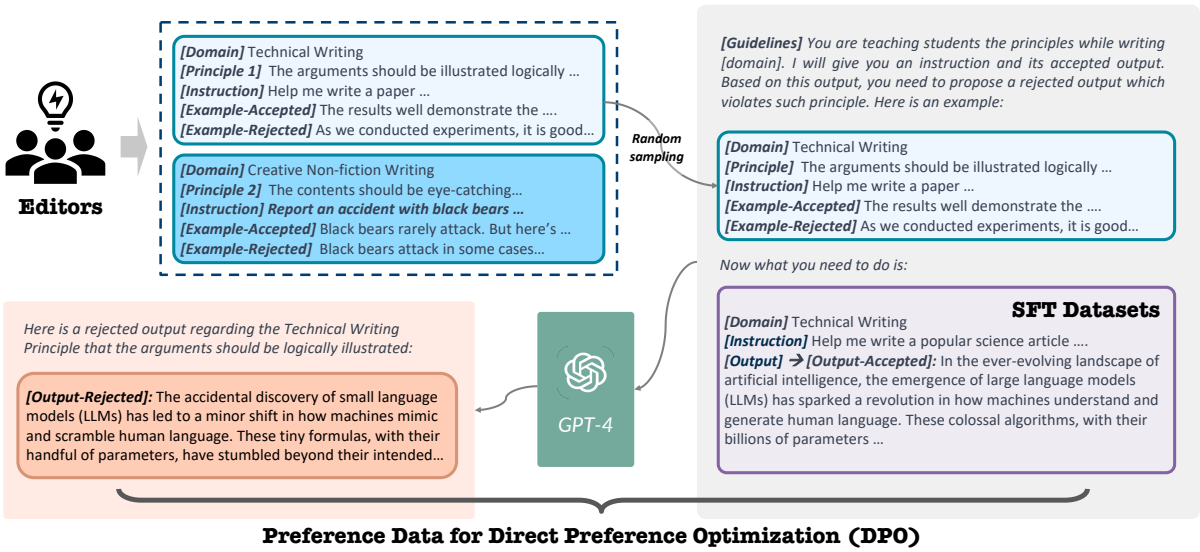

Figure 2 | Illustration of the Constitutional DPO framework.

3.3. Constitutional DPO: Learning From Principled Negative Examples

Finally, we propose Constitutional DPO, a novel alignment method that encourages LLMs to learn from preference data consisting of samples from the optimal policy and "principled" negative examples synthesized with AI feedback. Our approach combines the advantages of Constitutional AI (Bai et al., 2022; Sun et al., 2023), which train reward models based on principles written by human experts, RLCD (Yang et al., 2023a), which prompt LLMs to generate positive/negative examples and train reward models with AI-generated preference data, and DPO (Rafailov et al., 2023), which omits